

ICE 3107: Digital Signal Processing

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Lecture Outcomes

Analyze rational Z-transforms of discrete-time systems and determine poles and zeros to assess system behavior, and stability.

Convolution using Z-Transform

Convolution of two sequences

$$\text{If} \quad x_1(n) \leftrightarrow X_1(z)$$
$$x_2(n) \leftrightarrow X_2(z)$$

$$\text{then} \quad x(n) = x_1(n) * x_2(n) \leftrightarrow X(z) = X_1(z)\, X_2(z)$$

Convolution using Z-Transform (Cont.)

Example 3.2.9

Compute the convolution x(n) of the signals:

$$x_1(n) = \{1, -2, 1\}$$

$$x_2(n) = \begin{cases} 1, & 0 \le n \le 5 \\ 0, & \text{elsewhere} \end{cases}$$

Ans:

$$X(z) = X_1(z)X_2(z) = 1 - z^{-1} - z^{-6} + z^{-7}$$

$$x(n) = \{1, -1, 0, 0, 0, 0, -1, 1\}$$
$$\uparrow$$

Rational Z-Transform

What is a Rational Z-Transform?

A rational Z-transform is a ratio of two polynomials in z (or z^{-1}):

$$X(z)=\frac{N(z)}{D(Z)}$$

Why Rational Z-Transform is Important:

- Analyze systems
- Find stability
- Understand frequency behavior
- Design filters (FIR, IIR)

Rational Z-Transform (Cont.)

- **General Rational Form:**

$$X(z) = \frac{N(z)}{D(z)} = \frac{b_0 + b_1 z^{-1} + \cdots + b_M z^{-M}}{a_0 + a_1 z^{-1} + \cdots + a_N z^{-N}} = \frac{\sum_{k=0}^{M} b_k z^{-k}}{\sum_{k=0}^{N} a_k z^{-k}}$$

- **Factored Form:**

$$H(z) = K \frac{(z - z_1)(z - z_2) \cdots (z - z_M)}{(z - p_1)(z - p_2) \cdots (z - p_N)}$$

Poles and Zeros

A rational Z-transform means:

$$X(z) = \frac{N(z)}{D(z)}$$

Zero: Values of z that make numerator = 0: *N(z)=0*

Pole: Values of z that make denominator = 0: *D(z)=0*

Poles and Zeros (Cont.)

- Near zero → output becomes small
- Near pole → output becomes very large

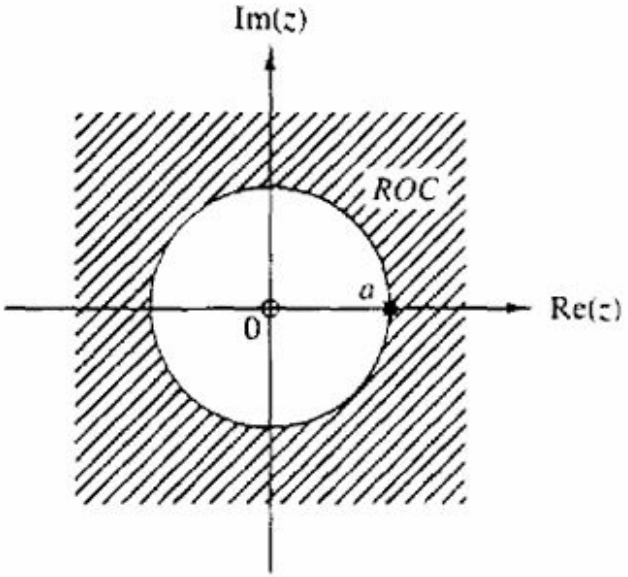
- Zero = “signal killer”
- Pole = “signal booster”

Poles and Zeros (Cont.)

Example 3.3.1 Determine the pole-zero plot for the signal $x(n)=a^n u(n)$, $a>0$

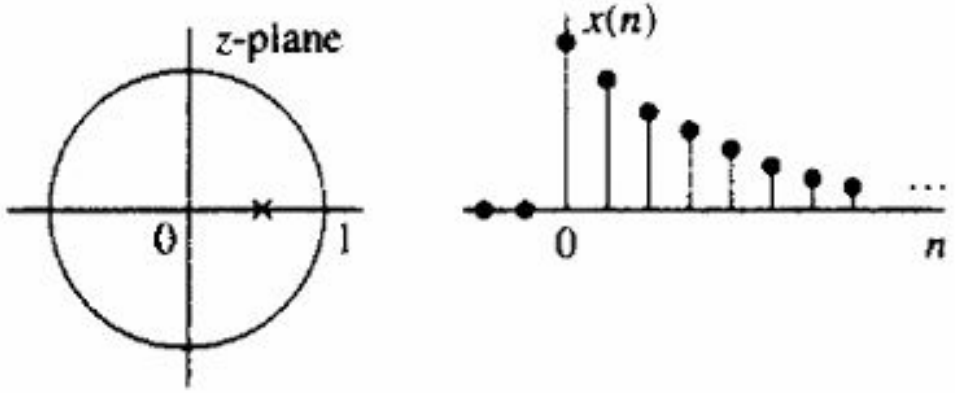
Ans:

$$X(z) = \frac{1}{1 - az^{-1}} = \frac{z}{z - a} \quad \text{ROC: } |z| > a$$



Thus X(z) has one zero at $z_1=0$ and one pole at $p_1=a$.

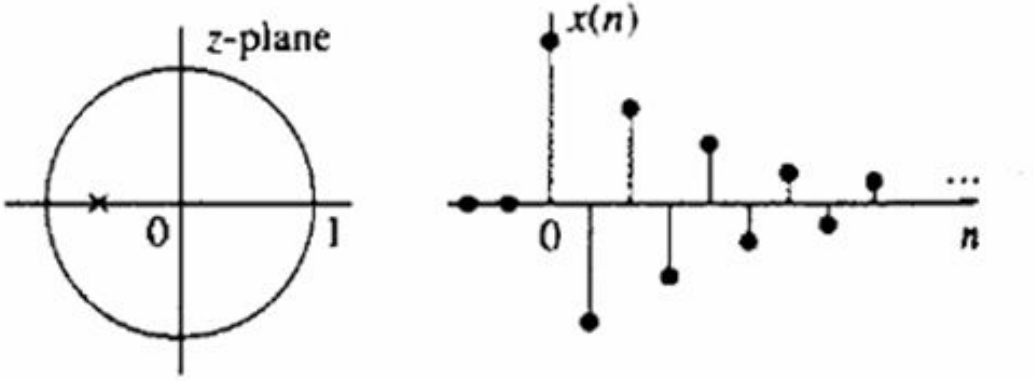
Pole Location and Time-Domain Behavior for Causal Signals



Positive pole inside unit circle:

- ✔ Pole located at $0<a<1$ → inside unit circle ⇒ $x(n)=a^n$ decays toward zero
- ✔ Time response is monotonically decreasing (no sign change, smooth exponential decay)

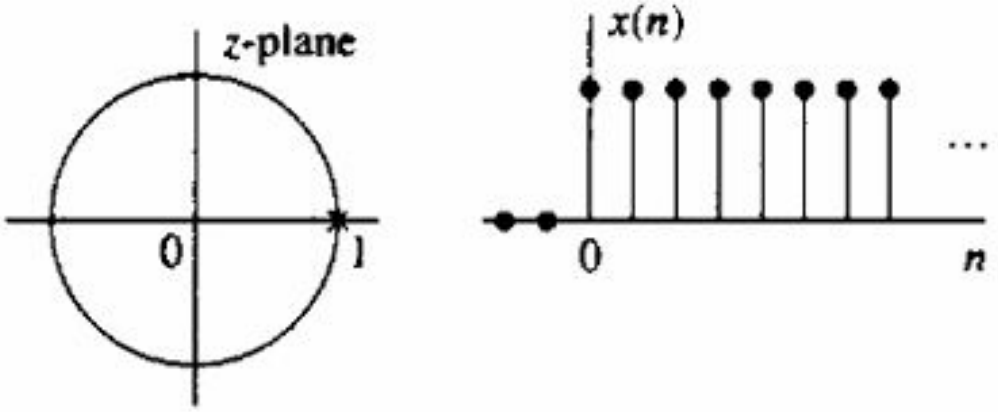
Pole Location and Time-Domain Behavior for Causal Signals (Cont.)



Negative pole inside unit circle

- ✔ Pole at $a=-r$, $|a|<1 \Rightarrow x(n) = (-r)^n = r^n(-1)^n$
- ✔ Signal decays with alternating signs (positive–negative oscillation)

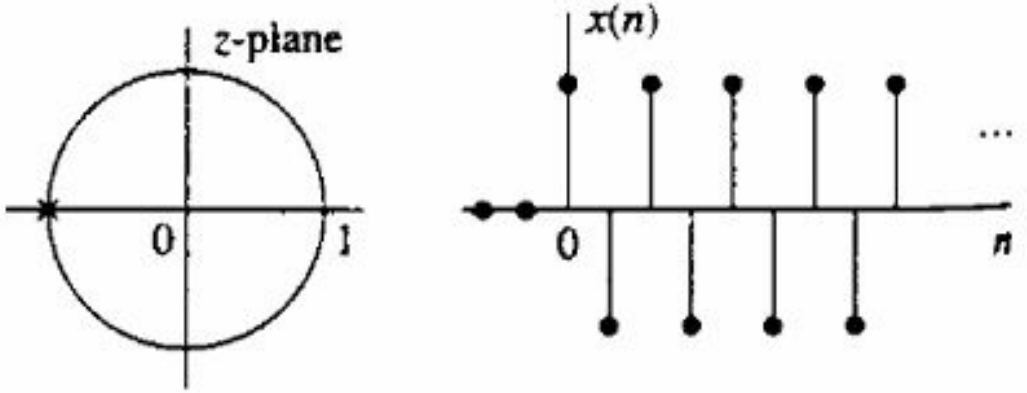
Pole Location and Time-Domain Behavior for Causal Signals (Cont.)



Pole ON unit circle, positive:

- ✔ Pole at a=1 ⇒ x(n)=1^n=1 (constant signal)
- ✔ Signal is non-decaying and non-growing → constant amplitude over time

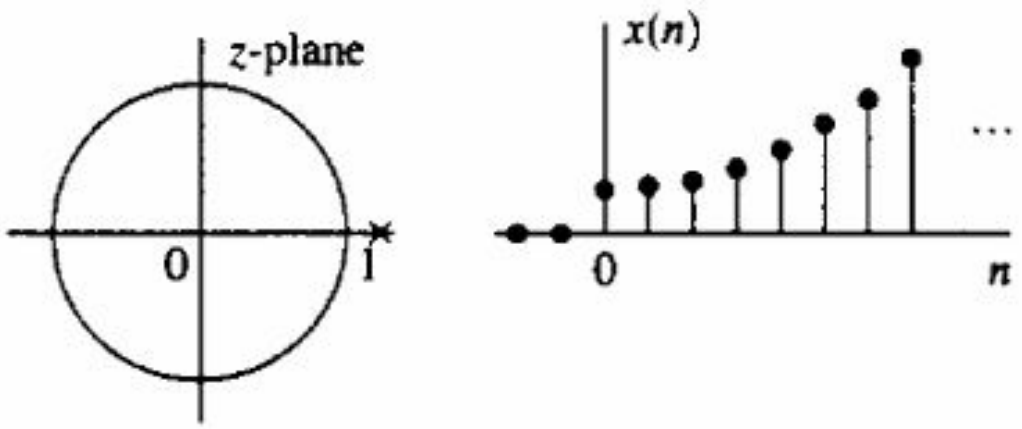
Pole Location and Time-Domain Behavior for Causal Signals (Cont.)



Pole ON unit circle, negative:

- ✔ Pole at $a=-1 \Rightarrow x(n)=(-1)^n$
- ✔ Signal oscillates between +1 and -1 (alternating, no decay)

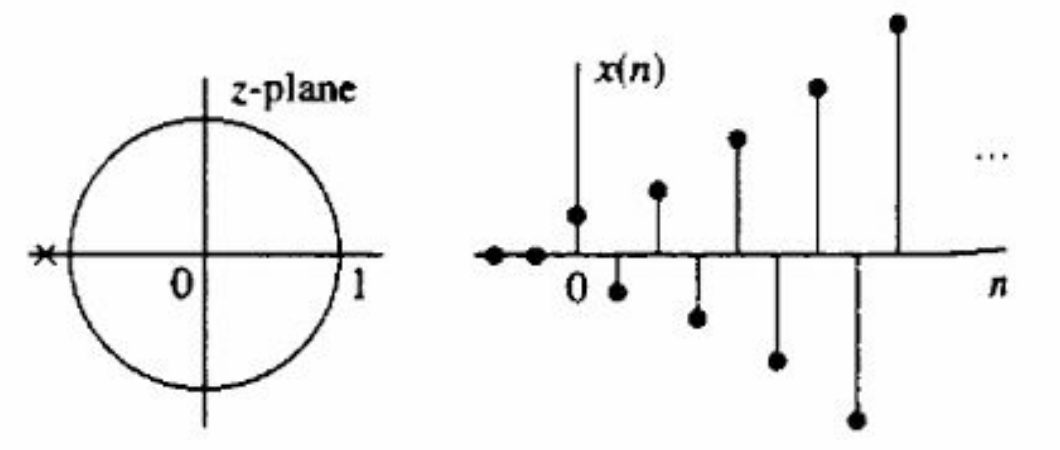
Pole Location and Time-Domain Behavior for Causal Signals (Cont.)



Pole OUTSIDE unit circle, positive:

- ✔ Pole at $a>1 \Rightarrow x(n)=a^n$ grows exponentially
- ✔ Signal is monotonically increasing (no sign change, unstable growth)

Pole Location and Time-Domain Behavior for Causal Signals (Cont.)



Pole OUTSIDE unit circle, negative:

- ✔ Pole at $a=-r$, $r>1 \Rightarrow x(n)=(-r)^n$
- ✔ Signal grows with alternating signs (oscillatory and unstable)

Practice Problem 01

$$H(z) = \frac{1}{1 + 0.8z^{-1}}$$

- Find zeros & poles
- Check stability
- Decide decaying / constant / growing / oscillatory
- Draw pole-zero plot

Answer:

✔ Zero: z=0

✔ Pole: z=−0.8

✔ Stability: Stable ($|p|<1$)

✔ Behavior: Decaying oscillatory

Practice Problem 02

$$H(z) = \frac{z + 0.9}{z^2 - 1.8z + 0.81}$$

- Find zeros & poles
- Check stability
- Decide decaying / constant / growing / oscillatory
- Draw pole-zero plot

Answer:

✔ Zero: z=−0.9

✔ Poles: z=0.9 (double pole)

✔ Stability: Stable (|p|<1)

✔ Behavior: Decaying, non-oscillatory

Practice Problem 03

$$H(z) = \frac{z + 0.5}{(z - 0.6)(z - 1.3)}$$

- Find zeros & poles
- Check stability
- Decide decaying / constant / growing / oscillatory
- Draw pole-zero plot

Answer:

✔ Zero: z=−0.5

✔ Poles: z=0.6,1.3

✔ Stability: Unstable

✔ Behavior: Growing (dominated by 1.3^n)

Thank You


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